

PILOTLESS HELICOPTER

UAV helicopters can perform a variety of functions, including communications, reconnaissance, and supply. Remote-controlled helicopters were first used by American forces in the 1960s. The Gruman MQ-8 Fire Scout flew in Libya in 2011 and the Kaman K-Max in Afghanistan in 2011.

itself fighting in the 21st century, with small groups of terrorists and guerrillas as the prime enemy. It was openly admitted that a dronelike Predator, with a maximum speed of 135mph (217kph), would not be fit to survive in combat against a modern state armed with an array of anti-aircraft weaponry and supersonic fighter aircraft.

Gradual progress

Developments in piloted military aircraft will probably not be startling in the foreseeable future. The crop of fighters that came into service early in the 21st century, including the F-22, the Eurofighter Typhoon, and the

Dassault Rafale, are likely to remain state-of-the-art for decades, while the F-35 fighter introduced in 2016 is planned to remain operational up to 2070. The B-52 bomber, in service since the mid-1950s, looks set to continue

on duty until close to its centenary. Similarly, in commercial aviation, no one appeared likely to rush to build an airliner bigger than the Airbus A380. Small passenger jets will probably reintroduce supersonic flight for a wealthy elite, but progress in mainstream commercial aviation is concentrating on reducing fuel and maintenance costs, and on cutting

emissions and reducing noise, a response to unrelenting criticism from environmentalists. Robotic passenger aircraft remain a future possibility, but nervous travelers are still likely to prefer the comforting presence of a pilot.

Human imagination is sometimes more limited than the human technological inventiveness. For example, flight pioneers of the Wright brothers' era were firmly convinced that aircraft would always be small; nothing could have amazed them more than the A380. What was achieved in the first century of flight was literally beyond imagination in the early 1900s. Along the way, a certain elated optimism associated with the early dream of flight has undoubtedly been sullied. When American aviation pioneer Octave Chanute wrote of the future of flight in 1894, he asked his readers to hope "that the advent of a successful flying machine... will bring nothing but good into the world; that it shall abridge distance, make all parts of the globe accessible, bring men into closer relation with each other, advance civilization, and hasten the promised era in which there shall be nothing but peace and goodwill among men." Part of this flight has been done, but



GLOBAL HAWK

A superstar among American UAVs, Global Hawk is a high-altitude reconnaissance aircraft. The basic model is 44ft (13.4m) long, with a wingspan of 116 ft (35.4m). Its maximum speed is over 350mph (600kph).

not the most optimistic part. Yet human flight remains a magical achievement in which imaginative aspiration and technology are welded, just as aesthetics and functionality have reached perfect accord in the greatest aircraft designs. Inevitably aircraft became a tool of business and war, but at the heart of aviation still lies the unquenched aspiration – as famously expressed by poet-pilot John Magee – to slip “the surly bonds of earth” and tread “the high, untrespassed sanctity of space.”